

Mechanics Problem Sheet

Student:
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Instructions

Attempt all questions. Show full working and include diagrams where necessary.

1. Motion in One Dimension

- Q1.** A car starts from rest and accelerates uniformly at 2 m/s^2 for 5 s. Find: (a) final velocity (b) distance travelled
- Q2.** A particle moves with initial velocity 10 m/s and decelerates at 2 m/s^2 . Find: (a) time to stop (b) total distance travelled
- Q3.** Two cars start from the same point: Car A moves with constant velocity 20 m/s . Car B starts from rest and accelerates at 4 m/s^2 . When does B overtake A?
- Q4.** A ball is thrown vertically upward at 15 m/s . Find: (a) maximum height (b) total time in air

2. Motion in Two Dimensions

- Q5.** A projectile is launched at 20 m/s at 30° . Find: (a) time of flight (b) maximum height
- Q6.** A projectile is launched at 25 m/s at 45° . Find the range.
- Q7.** A ball is thrown horizontally from a cliff 45 m high with speed 10 m/s . Find: (a) time to hit ground (b) horizontal distance
- Q8.** A particle has velocity $\vec{v} = 3\hat{i} + 4\hat{j} \text{ m/s}$. Find: (a) speed (b) direction (angle with horizontal)

3. Newton's Second Law and Free Body Diagrams

- Q9.** A 5 kg block is pulled with force 20 N on a frictionless surface. Find acceleration.
- Q10.** A 4 kg block is pulled with 16 N . Coefficient of friction $= 0.2$. Find acceleration.
- Q11.** A 6 kg block slides down a 30° incline (no friction). Find acceleration.

- Q12.** A 10 kg block is pushed horizontally with 50 N. Friction force = 20 N. Find acceleration.
- Q13.** Two blocks $m_1 = 2\text{ kg}$ and $m_2 = 3\text{ kg}$ are connected by a string on a frictionless surface. A force of 10 N is applied on m_1 . Find: (a) acceleration of the system (b) tension in the string

4. Newton's Third Law

- Q14.** A person pushes a wall with force 100 N. What force does the wall exert on the person?
- Q15.** A 5 kg object pushes a 3 kg object. The interaction force between them is 10 N. What force does each exert on the other?
- Q16.** A truck pulls a car with a rope. The tension in the rope is 500 N. Explain using Newton's Third Law what forces act on: (a) the truck (b) the car

5. Work, Energy, and Power

- Q17.** A force of 10 N moves a box 5 m in the same direction. Find the work done.
- Q18.** A 2 kg object moves at 6 m/s. Find kinetic energy.
- Q19.** A 3 kg object is pushed from rest by a constant force of 12 N over 4 m. Find final velocity.
- Q20.** A machine does 2000 J of work in 10 s. Find power.